Working with HTTP and FastAPI

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Overview

Custom response types

Serving a website

- Dynamic HTML with Jinja
- Processing Form Data

Status codes and error handling

Headers and cookies

Middleware

Organizing our code with APIRouter

Operation Parameters and HTTP

```
from fastapi import Form, Cookie, Headers, Request, Response

@app.get(...)
def my_operation(
    # Form Data
    input1: Annotated[int, Form()],
    input2: Annotated[str, Form()],

    # Cookies or Headers
    my_cookie: Annotated[str | None, Cookie()] = None,
    my_cookie: Annotated[str | None, Header()] = None,

    # Request and Response
    request: Request, response: Response):

    response.set_cookie(key="cookie_name", value="yummie!")
    response.headers["X-Pluralsight-Course-Header"] = "fastapi-course"
    response.status_code = 418
```

Operations in Multiple Files

routers/cars.py

```python
from fastapi import APIRouter


router = APIRouter(prefix="/api/cars")

@router.get
def get_cars():
    ...
```

carsharing.py

```python
from routers import cars


app = FastAPI(title="Car Sharing")
app.include_router(cars.router)
```

Serving HTML

```
from starlette.responses import HTMLResponse


@router.get("/", response_class=HTMLResponse)
def home():
    return “<html><body><h1>Welcome!</h1></body></html>”
```

Handling Exceptions

```
from starlette.responses import JSONResponse


@app.exception_handler(SomeExceptionType)
async def badtrip_exception_handler(request: Request, exc: SomeExceptionType):
    return JSONResponse(
        content={"message": ”Something went wrong”},
    )
```

Middleware

```python
@app.middleware("http")
async def my_middleware(request: Request, call_next):
    # do something with request
    log_request(request)

    # let FastAPI handle request
    response = await call_next(request)

    # do something with response
    change_response(response)
    return response
```

Summary

- Organizing our code with APIRouter
- Custom response types
- Serving a website
- Status codes and error handling
- Headers and cookies
- Middleware

Up Next: Authentication